

Meta-Learning for Few-Shot Personalization of Error-Related Potential Brain-Computer Interfaces

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Abstract—When a brain-computer interface (BCI) errs, the user’s electroencephalogram carries an error-related potential (ErrP) the system can use to self-correct, but exploiting it usually costs every new user tens of minutes of calibration. We ask whether meta-learning can replace that calibration with four labeled trials, and answer with a controlled benchmark on two structurally different ErrP paradigms under leave-one-subject-out evaluation with subject-level statistics ($N=16$ each). Gradient meta-learners and a FiLM-conditioned encoder on principal-component features beat a from-scratch decoder by 5–7 points on one corpus and 15–16 on the other, and the gain survives realistic calibration: with four consecutive, unbalanced trials as a user would supply them, 15–16 of 16 subjects still improve. Within that representation, class-weighting the episodic loss helps full MAML (+1.9–2.9 points) but not head-only ANIL. A matched comparison, however, changes the conclusion. An EEGNet pretrained on the source subjects’ raw epochs and applied with no calibration at all beats the best principal-component meta-learner by about six points on both corpora (13–15 of 16 subjects), and adapting it from four trials, whether by meta-learning or by fine-tuning, adds no significant gain. In these data, cross-subject pretraining on an adequate input representation, not the meta-learning objective, carries few-shot ErrP personalization.

Index Terms—brain-computer interface, error-related potential, meta-learning, few-shot learning, EEG, personalization

I. INTRODUCTION

A user who cannot speak or move spells a word letter by letter through a brain-computer interface (BCI), and the speller picks the wrong letter. Their brain has already registered the mistake: within roughly 250ms of the erroneous feedback the EEG carries a stereotyped *error-related potential* (ErrP), a fronto-central negativity followed by a positivity near 320ms [1]. A decoder that reads that signal lets the interface undo its own mistake, with no extra effort from the user [2].

Calibration is what stands in the way. ErrP morphology shifts with anatomy, electrode placement, and signal quality, so a decoder trained on one person transfers poorly to the next, and every new user usually sits through tens of minutes of labeled trials first, a large part of why ErrP-BCIs rarely leave the lab [3]. Meta-learning offers a way around it: a model trained across many users learns an initialization that adapts

to a new user from a handful of trials [4]. Whether that works for single-trial ErrP decoding, and against which alternatives, is the question we answer.

Novelty and objective. Meta-learning for EEG has targeted mostly motor imagery [5]–[7], while ErrP decoding has advanced through deeper supervised or transfer models trained on substantial labeled data [3], [8] or through online reinforcement learning [9]. To our knowledge, no prior work tests few-trial ErrP personalization across more than one paradigm with subject-level statistics. Our objective is not a new algorithm but a controlled answer to whether, and why, meta-learning helps. We contribute:

- **a four-trial ErrP personalization benchmark** of six meta-learning methods against source-aware baselines and matched controls under one leave-one-subject-out (LOSO) protocol, replicated on a second paradigm: meta-learners beat a from-scratch decoder by 5–7 (INRIA) and 15–16 points (coadaptation), also when calibration is realistic (unbalanced, consecutive trials, single draws);
- **a matched analysis of input representation and baseline strength**, showing that an unadapted source-pretrained EEGNet on raw epochs outperforms the principal-component meta-learners and that four-trial adaptation adds nothing significant to it; and
- **ablations and practical measures**: class-weighting the episodic loss helps full MAML but not head-only ANIL, and we report calibration burden, error-correction rates, and compute cost.

II. RELATED WORK

ErrP detection. ErrPs serve both to correct BCI output and as reward signals for autonomous agents [1], [2]. Single-trial detection remains hard; surveys of linear, kernel, and Riemannian classifiers point to preprocessing and heavy class imbalance as the recurring issues [3], and recent work adds physiological covariates [10]. EEGNet is a widely used compact neural decoder [11].

Cross-subject transfer. Beyond meta-learning, a major line is domain adaptation: Riemannian and Euclidean alignment before a shared classifier [12], [13], and, for ERPs, minimum distance to (weighted) class means on xDAWN super-

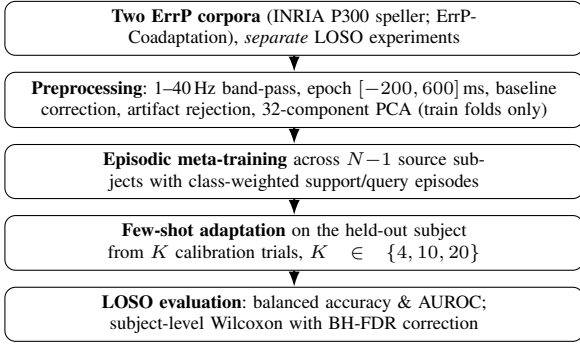


Fig. 1. Overview of the few-shot ErrP personalization pipeline.

trial covariances [14], [15]. EEG foundation models pursue calibration-free use at far greater data and compute cost [16].

Meta-learning. MAML learns an initialization that adapts in a few gradient steps [4]; ANIL adapts only the head [17]; Reptile uses first-order updates [18]; Prototypical [19] and Matching networks [20] classify by similarity in a learned embedding; FiLM conditions a network on side information [21]. For ErrPs, a recent CNN–transformer with transfer learning reports 71.8% leave-one-subject-out and 78.7% within-session accuracy [8]; few-trial personalization has received far less attention.

III. METHOD

A. Problem formulation

A decoder is prepared on $N-1$ source subjects and specialized to a new user from K of that user’s own trials; under LOSO each subject is the new user once and never contributes to the model tested on it. ErrP data is imbalanced (error prior ~ 0.32 INRIA, ~ 0.28 coadaptation), so the main benchmark draws a *class-balanced* calibration set of $\lfloor K/2 \rfloor$ trials per class; the regime of interest is $K=4$ (two error, two correct trials), and we sweep $K \in \{4, 10, 20\}$. Sec. V relaxes the balancing.

B. Personalization pipeline

Fig. 1 shows the pipeline: each corpus is band-pass filtered, epoched, reduced to 32 whitened principal components, and turned into class-balanced episodes; a meta-model is trained across the source subjects under a class-weighted episodic loss and adapted to the held-out subject in five gradient steps from its K calibration trials.

C. Datasets and preprocessing

The **primary** corpus is the INRIA BCI Challenge (NER 2015) P300 speller (16 subjects, feedback-locked ErrP) [22]; the **second** is ErrP-Coadaptation [23] (all 16 released subjects, S03–S18), an observation-ErrP paradigm in which a participant infers a robot’s intended object from its gaze. They differ in task, montage, and sampling rate (200 vs. 250 Hz); we run LOSO *separately* on each and never train on one and test on the other. A 4th-order zero-phase Butterworth band-pass (1–40 Hz; plus a 50 Hz notch at 250 Hz) precedes

TABLE I
METHODS. SRC.: SEES SOURCE SUBJECTS; CW: CLASS-WEIGHTED LOSS; *: ADDED IN THE MATCHED BENCHMARK (SEC. V).

Method	Src.	CW	Test-time adaptation	Input
Full-MAML / Reptile	yes	yes	gradient, full (5 steps)	PCA
MAML-ANIL	yes	yes	gradient, head (5 steps)	PCA
Subj.-Cond.	yes	yes	FiLM + gradient, head	PCA
Prototypical / Matching	yes	bal. ep.	prototype / attention	flat
EEGNet (probe)	yes	yes	LR probe (frozen)	epochs
EEGNet-ZeroShot*	yes	yes	none	epochs
EEGNet-FT*	yes	yes	all layers (50 steps)	epochs
ANIL-EEGNet*	yes	yes	gradient, head (5 steps)	epochs
Reptile-EEGNet*	yes	yes	gradient, full (5 steps)	epochs
Riemannian / Cov. Align.	yes	LDA	LDA on K	epochs
xDAWN-MDM* / MDWM*	yes	–	class means (MDWM: blended)	epochs
Supervised	no	yes	train from scratch	PCA
Pretrain-FT	yes	yes	LR probe (frozen)	PCA
Pretrain-ZeroShot	yes	yes	none	PCA

epoching; epochs span $[-200, 600]$ ms around feedback, are baseline-corrected over $[-200, 0]$ ms, and are rejected above $\pm 100 \mu\text{V}$. The meta-learners use 32 whitened principal components of the flattened waveform (PCA fit per fold on training subjects only); EEGNet and the covariance methods use raw epochs. Sec. V tests what this asymmetry costs.

D. Methods, baselines, and controls

Six **personalization methods** run in the pipeline: Full-MAML [4], MAML-ANIL [17], and Reptile [18] (gradient-based); Prototypical [19] and Matching [20] networks (metric-based); and **SubjectConditioned**, a FiLM encoder [21] that computes error/correct prototypes from the support set and modulates a feature extractor. All episodic losses are class-weighted by inverse class frequency.

From-scratch reference. Supervised, a three-layer MLP (128-64-2, LayerNorm, GELU, dropout 0.3) trained on the K trials alone, the only method that never sees source data. **Matched controls** see the same source subjects with the same class weighting as the meta-learners: Pretrain-FT freezes a source-pretrained MLP and fits a logistic-regression probe on the K trials; Pretrain-ZeroShot keeps its source head and does not adapt. **Source-aware baselines:** EEGNet [11] pretrained on the source subjects (class-weighted cross-entropy), used frozen with an ℓ_2 logistic probe on the K support embeddings, as in the main benchmark; and xDAWN/tangent-space Riemannian [24] and covariance-alignment classifiers [12] whose LDA is fit on the K trials. Sec. V adds the raw-input variants of Table I.

E. Implementation details

MAML uses inner LR 0.01 (5 steps) and outer LR 5×10^{-4} (Adam, cosine decay, gradient clipping at 1); Full-MAML is second-order, ANIL adapts only the head (first-order). Reptile uses 5 inner SGD steps and outer step 0.1. SubjectConditioned derives FiLM γ/β from a 32-dim subject embedding. Meta-training runs 2000 iterations (meta-batch 4 subjects; episodes of 10 support + 40 query, class-balanced). EEGNet ($F_1=8$, $D=2$, $F_2=16$, temporal kernel $f_s/2$) is pretrained

TABLE II
PRIMARY DATASET (INRIA): BALANCED ACCURACY (MEAN $\pm$ STD, 3 SEEDS, $N=16$) AND AUROC AT $K=4$.

Method	$K=4$	$K=10$	$K=20$	AUROC
EEGNet (probe) [†]	0.600 $\pm$.009	0.628 $\pm$.004	0.645 $\pm$.011	0.640
Reptile ^{†‡}	0.594 $\pm$.002	0.600 $\pm$.003	0.607 $\pm$.003	0.637
Subj.-Cond. ^{†‡}	0.584 $\pm$.007	0.584 $\pm$.009	0.587 $\pm$.013	0.617
Full-MAML ^{†‡}	0.584 $\pm$.009	0.590 $\pm$.011	0.590 $\pm$.008	0.625
MAML-ANIL ^{†‡}	0.581 $\pm$.013	0.580 $\pm$.015	0.585 $\pm$.015	0.614
Prototypical	0.536 $\pm$.005	0.553 $\pm$.009	0.569 $\pm$.002	0.548
Matching	0.535 $\pm$.002	0.527 $\pm$.003	0.542 $\pm$.022	0.546
Cov. Align.	0.526 $\pm$.026	0.546 $\pm$.025	0.562 $\pm$.020	0.524
Riemannian	0.517 $\pm$.015	0.561 $\pm$.026	0.541 $\pm$.040	0.542
Pretrain-ZeroShot	0.563 $\pm$.007	0.563 $\pm$.007	0.563 $\pm$.007	0.601
Pretrain-FT	0.537 $\pm$.014	0.542 $\pm$.012	0.544 $\pm$.018	0.563
Supervised	0.527 $\pm$.034	0.563 $\pm$.015	0.577 $\pm$.004	0.538

with Adam (10^{-3}) and early stopping; at $K < 10$ its probe additionally sees three mixup interpolations per support trial. These hyperparameters are based on the cited papers’ defaults and were fixed on INRIA before the LOSO evaluation, using one pilot subject; they were then applied unchanged to the coadaptation corpus, which thus checks against dataset-specific tuning. The matched-benchmark settings were fixed a priori and never tuned: EEGNet-FT fine-tunes all layers (Adam 10^{-4} , 50 steps, BatchNorm statistics frozen); ANIL-/Reptile-EEGNet reuse the MAML/Reptile settings on an EEGNet backbone; MDWM uses trade-off 0.5 on 4-filter xDAWN super-trial covariances. Code and per-fold results are released: <https://github.com/Md-Shahidul-Islam-Dipu/Errp-Meta>.

F. Protocol and statistics

The main benchmark uses three seeds (42, 123, 456) and averages the episodic methods and EEGNet over 10 calibration draws (other baselines: one draw). The unit of inference is the *subject*: we average each subject over seeds and draws and run paired Wilcoxon signed-rank tests at $N=16$ (exact null; floor $p=3.1 \times 10^{-5}$), with Benjamini–Hochberg FDR correction within each family of comparisons (69 per dataset in the main benchmark) and rank-biserial effect size r . For headline pairs we give the median paired difference with a 95% bootstrap CI over subjects. Table $\pm$ values are across-seed spread. We report balanced accuracy and AUROC, both robust to imbalance.

IV. MAIN BENCHMARK

A. Primary dataset

In Tables II–III, [†] marks methods that significantly beat Supervised at $K=4$ (BH-FDR $p < 0.05$) and, in Table II, [‡] methods not significantly different from EEGNet (probe) at $K=4$; bold marks the best entry per column.

At $K=4$ the gradient meta-learners, the FiLM encoder, and EEGNet (probe) significantly outperform Supervised (Table II; $r = 0.82$ – 1.0 , $\Delta = 5$ – 7 pp; Reptile median $+0.051$, 95% CI $[+0.038, +0.086]$), with 15 of 16 subjects improving under Reptile. No meta-learner differs significantly from EEGNet

TABLE III
SECOND DATASET (ERRP-COADAPTATION); FORMAT AS TABLE II.

Method	$K=4$	$K=10$	$K=20$	AUROC
Subj.-Cond. [†]	0.721 $\pm$.006	0.724 $\pm$.004	0.723 $\pm$.008	0.800
Reptile [†]	0.720 $\pm$.003	0.724 $\pm$.005	0.731 $\pm$.006	0.799
MAML-ANIL [†]	0.717 $\pm$.001	0.720 $\pm$.001	0.722 $\pm$.001	0.799
Full-MAML [†]	0.716 $\pm$.003	0.720 $\pm$.001	0.724 $\pm$.001	0.801
Prototypical [†]	0.681 $\pm$.007	0.727 $\pm$.002	0.732 $\pm$.008	0.744
EEGNet (probe) [†]	0.654 $\pm$.004	0.713 $\pm$.009	0.749 $\pm$.003	0.734
Matching [†]	0.650 $\pm$.009	0.674 $\pm$.006	0.678 $\pm$.013	0.692
Pretrain-ZeroShot [†]	0.700 $\pm$.001	0.700 $\pm$.001	0.700 $\pm$.001	0.767
Pretrain-FT	0.590 $\pm$.021	0.643 $\pm$.013	0.653 $\pm$.014	0.649
Riemannian	0.514 $\pm$.007	0.566 $\pm$.014	0.584 $\pm$.007	0.531
Cov. Align.	0.521 $\pm$.030	0.566 $\pm$.009	0.564 $\pm$.009	0.534
Supervised	0.563 $\pm$.013	0.581 $\pm$.018	0.614 $\pm$.011	0.593

(probe) at $K=4$ ($p = 0.49$ – 0.94); by $K=20$ EEGNet (probe) significantly exceeds Full-MAML, MAML-ANIL, and the FiLM encoder (BH-FDR $p < 0.02$). The metric-based and covariance methods do not beat Supervised at any K .

B. Replication on a second dataset

On the coadaptation paradigm all meta-learners beat Supervised at $K=4$ by 15–16 points ($p = 3.1 \times 10^{-5}$, $r = 1.0$, 16 of 16 subjects; Reptile median $+0.159$, CI $[+0.126, +0.187]$) and also beat EEGNet (probe) (Reptile $+0.065$, CI $[+0.035, +0.086]$; BH-FDR $p < 0.001$), a gap that closes by $K=10$. Across-seed spread is 0.001–0.008.

Objective vs. data access. Pretrain-FT and Pretrain-ZeroShot see the same source subjects with the same class weighting, so they separate the meta-learning objective from data access *within the PCA representation*. The meta-learners beat both on both corpora at every K (e.g., Reptile over Pretrain-FT: $+5.7$ INRIA, $+13.0$ coadaptation; over Pretrain-ZeroShot: $+3.1$ and $+2.0$; BH-FDR $p < 0.05$). Naive probe fitting on four noisy trials is itself harmful: Pretrain-FT trails its own zero-shot head. The next section asks whether these conclusions survive a matched input representation.

V. MATCHED BENCHMARK

The main benchmark leaves three confounds: meta-learners and EEGNet saw different inputs, EEGNet was only probed, and balanced, draw-averaged calibration is not what a user supplies. We therefore retrained the gradient meta-learners, the FiLM encoder, the controls, and EEGNet once per fold (seed 42), added the variants marked in Table I, and scored all of them on *identical* calibration draws under three protocols: *balanced* (as above), *natural* (K random trials, errors at the user’s own rate), and *contiguous* (K consecutive trials; draw 0 is the first K trials of the recording). Each method is scored on all remaining trials of the subject. When a draw holds a single class, gradient methods adapt on it, while probe and class-mean methods fall back to their source model. Table IV closely reproduces the main benchmark (Reptile 0.593 vs. 0.594 on INRIA, 0.731 vs. 0.720 on coadaptation); remaining differences, e.g. for EEGNet (probe), reflect the query set (all remaining trials rather than a class-balanced subset).

TABLE IV
MATCHED BENCHMARK AT $K=4$ (SEED 42, $N=16$): BALANCED
ACCURACY UNDER BALANCED (BAL.) AND CONTIGUOUS (CONT.)
CALIBRATION, AND AUROC (BALANCED).

Method	INRIA			Coadaptation		
	Bal.	Cont.	AUROC	Bal.	Cont.	AUROC
<i>Raw epochs, EEGNet backbone</i>						
ANIL-EEGNet	.671	.672	.738	.788	.789	.879
EEGNet-ZeroShot	.657	.657	.721	.792	.792	.876
EEGNet-FT	.649	.647	.707	.795	.789	.876
EEGNet (probe)	.577	.597	.618	.656	.677	.737
Reptile-EEGNet	.590	.567	.623	.673	.655	.737
<i>32-PCA features</i>						
Reptile	.593	.582	.628	.731	.717	.809
Subj.-Cond.	.581	.580	.617	.728	.722	.804
Full-MAML	.589	.554	.620	.723	.711	.803
MAML-ANIL	.573	.559	.605	.720	.714	.801
Pretrain-ZeroShot	.567	.568	.592	.702	.701	.774
Pretrain-FT	.525	.539	.540	.590	.618	.647
<i>xDAWN covariances</i>						
MDWM	.583	.567	.617	.642	.623	.713
xDAWN-MDM	.574	.554	.593	.612	.588	.664
Supervised	.531	.514	.546	.553	.524	.575

Input representation and baseline strength. The confound is large. The same ANIL learner gains 9.8 (INRIA) and 6.8 points (coadaptation) when moved from 32-PCA features to an EEGNet backbone on raw epochs (0/16 subjects favor PCA, $p = 3.1 \times 10^{-5}$). More consequentially, the source-pretrained EEGNet *with no calibration at all* (EEGNet-ZeroShot) beats the PCA meta-learners; over the best, Reptile, by +6.4 points on INRIA (13/16 subjects, BH-FDR $p = 0.005$) and +6.2 on coadaptation (15/16, $p < 0.001$). The main benchmark’s EEGNet looked weaker only because its four-trial probe discards the source classifier (EEGNet-FT beats it by 7.2 and 13.9 points, 16/16). Adapting the strong source model from four trials adds nothing significant: neither ANIL-EEGNet (+1.4, −0.4 points; BH-FDR $p = 0.22$, 0.50) nor full fine-tuning (−0.8, +0.3; uncorrected $p \geq 0.38$) beats EEGNet-ZeroShot at $K=4$; fine-tuning helps only marginally by $K=20$ (+0.9 points on coadaptation, uncorrected $p=0.004$). Raw input alone is not sufficient either: first-order Reptile on the EEGNet backbone does no better than on PCA. The Riemannian comparators do not beat Reptile on INRIA (BH-FDR $p \geq 0.44$) and trail it by 9–12 points on coadaptation (16/16).

Realistic calibration. Relaxing the balancing leaves the ranking intact. Under the contiguous protocol Reptile still beats Supervised by 6.8 (INRIA, 15/16, $p = 2 \times 10^{-4}$) and 19.3 points (coadaptation, 16/16, $p = 3.1 \times 10^{-5}$), and on the *same* draw it wins 84% and 100% of the time; its worst-decile draw (0.50, 0.64) is at or above Supervised’s (0.48, 0.49). Unbalanced draws matter: 28–36% of four-trial natural or contiguous draws contain no error trial at all. Measured on the recordings, collecting two error and two correct trials takes a median of 6 (INRIA) and 7 (coadaptation) raw trials, and 11 and 16 to cover 90% of starting points (five of each: medians

14 and 18.5).

Practical significance and cost. At the default decision threshold and the coadaptation error rate (28%), per 100 selections EEGNet-ZeroShot catches 22.2 of 28.4 errors while flagging 14.3 correct selections, Reptile catches 20.3 and flags 18.4, and Supervised catches 15.9 and flags 32.7. On INRIA (32% errors) every method flags about as many correct selections as it catches errors (best: ANIL-EEGNet, 18.8 caught, 17.2 flagged), so none is yet useful for automatic undo there. Offline, one LOSO fold (15 source subjects) costs 1.0–1.8 min of Tesla T4 time for EEGNet pretraining, 1.4–2.6 min for ANIL, Reptile, ANIL-EEGNet and the FiLM encoder, and 4.7–5.1 min for second-order Full-MAML; the full matched benchmark took 4.4 and 4.8 GPU-hours. Per-user adaptation takes 5–17 ms for the meta-learners, about 0.2 s for EEGNet fine-tuning, and none for EEGNet-ZeroShot.

VI. ABLATIONS

Fig. 2 isolates three design choices of the PCA pipeline on INRIA.

Class weighting of the episodic loss by inverse class frequency lifts Full-MAML by 1.9 and 2.9 points at $K=4$ and 10 (paired Wilcoxon $p=0.006$, 0.004; +2.1, $p=0.07$ at $K=20$) but leaves MAML-ANIL unchanged ($|\Delta| \leq 0.4$ points, $p \geq 0.86$), possibly because ANIL adapts only the head. **FiLM** gives a small significant gain at $K=4$ (+1.5 pp; $p=0.009$, $r=0.65$) that disappears by $K=10$ ($p > 0.27$). **PCA rank**: quadrupling it to 128 components did not help, which, together with the raw-input results, suggests that the limit lies in the flattened projection rather than in the number of components.

VII. DISCUSSION

Within a fixed PCA representation, meta-learning is effective: it beats a from-scratch decoder and controls matched on data access and class weighting, and class-weighting the episodic loss helps its full-network variant. It does not, however, beat a conventionally pretrained CNN that sees the raw epochs. Once that model’s source head is kept rather than replaced by a four-trial probe, it is the strongest calibration-free decoder we tested, and four trials of adaptation, meta-learned or fine-tuned, do not improve it. What carries few-shot ErrP personalization in our data is cross-subject pretraining on an adequate input representation.

Clinical implications. In the main benchmark the meta-learners’ edge over EEGNet was confined to four trials and closed by ten; the matched benchmark shows that this edge reflected the weak probe baseline rather than an advantage of meta-learning. The practical message is nonetheless favorable: a source-pretrained decoder is usable from the first trial of a session, which matters for users who tire quickly, and short per-session recalibration adds little. Where accuracy is still too low for automatic undo (INRIA), progress is more likely to come from choosing the operating point and from

Ablation studies (primary dataset, 3 seeds)

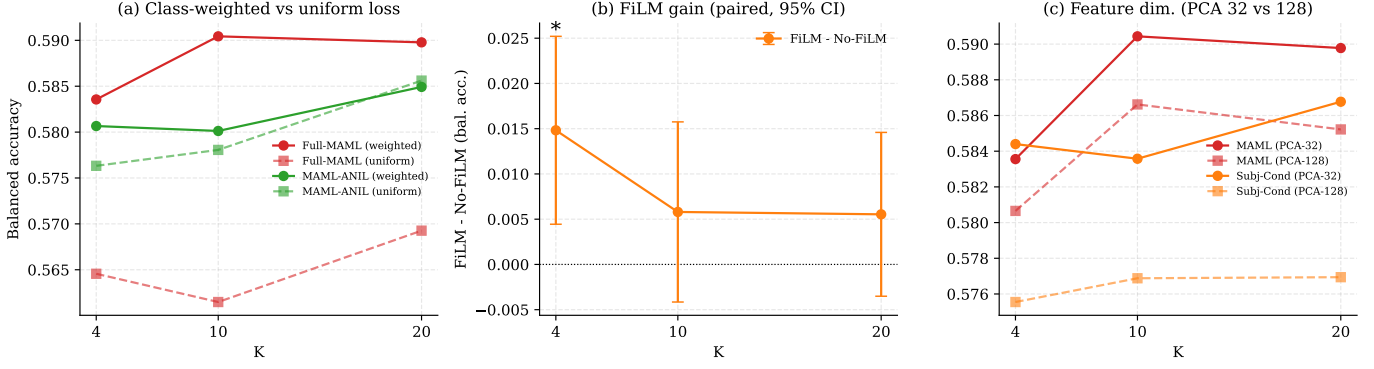


Fig. 2. Ablations (INRIA, 3 seeds). (a) class-weighted vs. uniform loss; (b) paired per-subject FiLM gain (FiLM – No-FiLM) with 95% CI, which excludes zero only at $K=4$ (*); (c) 32 vs. 128 PCA components.

better source models than from faster adaptation. A meta-learned initialization remains a reasonable choice when only compressed features are available.

Comparison with existing solutions. A recent ErrP decoder reports roughly 72–79% accuracy across leave-one-subject-out and within-session settings [8], and the single-trial classifiers surveyed in [3] fall in a comparable band; our calibration-free source EEGNet reaches 0.79 balanced accuracy on coadaptation and 0.66 on the harder INRIA speller. The comparison is indicative only: datasets differ, and balanced accuracy is stricter than raw accuracy.

Limitations. (i) The matched benchmark uses one seed; the main benchmark’s seed spread (≤ 0.013) suggests this is minor. (ii) The added EEGNet variants were not tuned, so they are, if anything, understated. (iii) Results are at the default threshold; error-recall at low false-alarm rates is not analyzed. (iv) All experiments are offline, on two datasets, with calibration and test trials from the same recording. (v) The pipeline’s hyperparameters were fixed with one INRIA pilot subject; the unchanged replication on coadaptation limits, but does not exclude, dataset-specific tuning.

VIII. CONCLUSION

We asked whether meta-learning can personalize an ErrP decoder from four calibration trials, on two paradigms with subject-level statistics. Meta-learners on principal-component features beat a from-scratch decoder and matched transfer controls, robustly and under realistic calibration, and class-weighting the episodic loss helps full MAML. A matched comparison shows that a source-pretrained EEGNet on raw epochs, with no calibration, beats them on both paradigms and is not improved by four-trial adaptation. Future work should combine strong raw-input source models with operating-point selection and validate them in closed-loop online use.

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